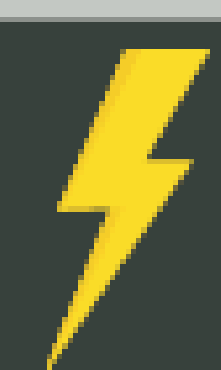


FLUORIDE-FREE SYNTHESIS OF A NOVEL MOF/MXENE HYBRID FOR HIGH PERFORMANCE SODIUM METAL BATTERIES

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1 Problem



Low Ionic Conductivity in Polymers



Flammable Liquid Electrolytes



Toxic MXene Synthesis

2 Objectives

Develop a fluoride-free MXene/MOF hybrid electrolyte for SMBs

Make highly stable electrode/electrolyte interface

Enhance the performance of SMB in wide temperature range

3 Introduction

Combining MXenes and MOFs in **polymer electrolytes** enhances ionic conductivity and stability. **MXenes** improve ion transport and interface stability, while **MOFs** support sodium-ion movement. Using a **fluoride-free molten salt** method makes MXene synthesis safer, making the overall system both high-performing and less toxic.

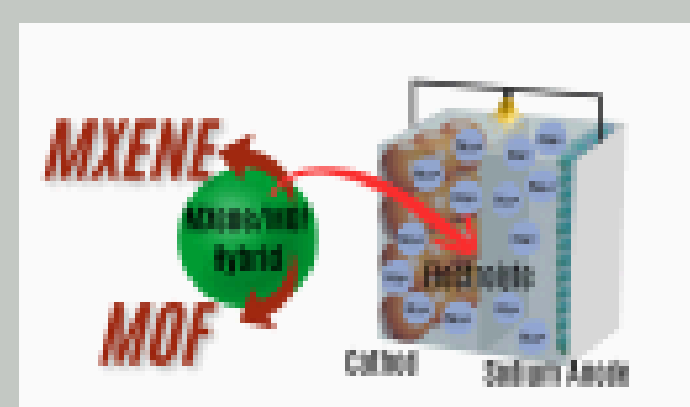


Fig. 1. Synergistic MXene-MOF Hybrid Material for enhanced SMB performance

Table 1. Comparison of MOF, MXene, and Hybrid properties highlighting their complementary features

MOFs	MXene	Hybrid
High Surface Area	Electronic Conductivity	Interlinked Structure
High Chemical Stability	Hydrophilicity	Large Active Layer
Poor Stability	Exfoliation Requirement	Enhanced Conductivity
Low Conductivity	Salt Reducing	Improved Stability

4 Novelty

The novelty is the **MXene/MOF filler** and using **MXene** as an **interface layer** for improved ionic conductivity and stability.

5 Methodology

Materials



Ti₃AlC₂ (MAX Phase): Titanium aluminum carbide powder was obtained from Sigma-Aldrich.



ZnCl₂ (LAMT): Zinc chloride was used as the Lewis acid molten salt, obtained from Riedel-de Haën.

1

Powder Mixing



Mixed Ti₃AlC₂ and ZnCl₂, then placed in Quartz Boat



2

Sintering

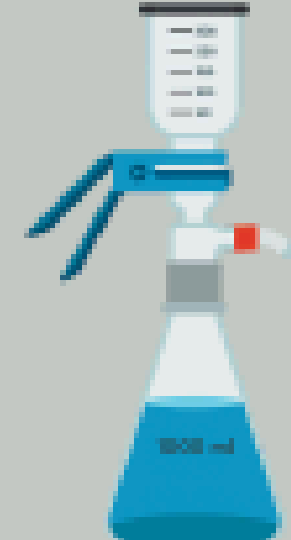


Heated at 650 °C under Nitrogen atmosphere

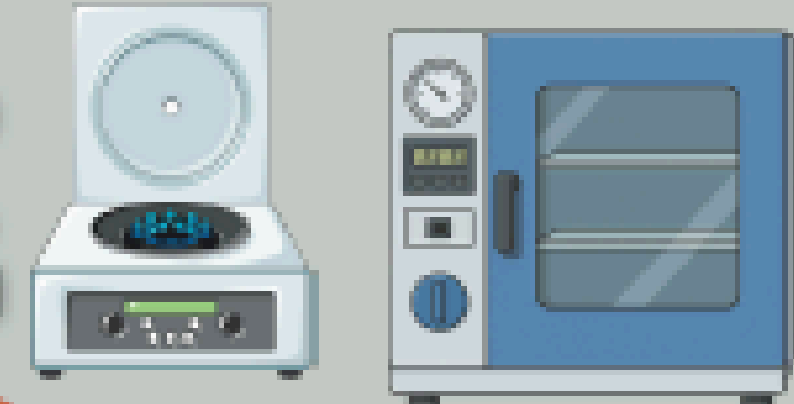


3

Washing and Drying



Etched, rinsed with water, centrifuged, then vacuum dried



6

6 Graphs

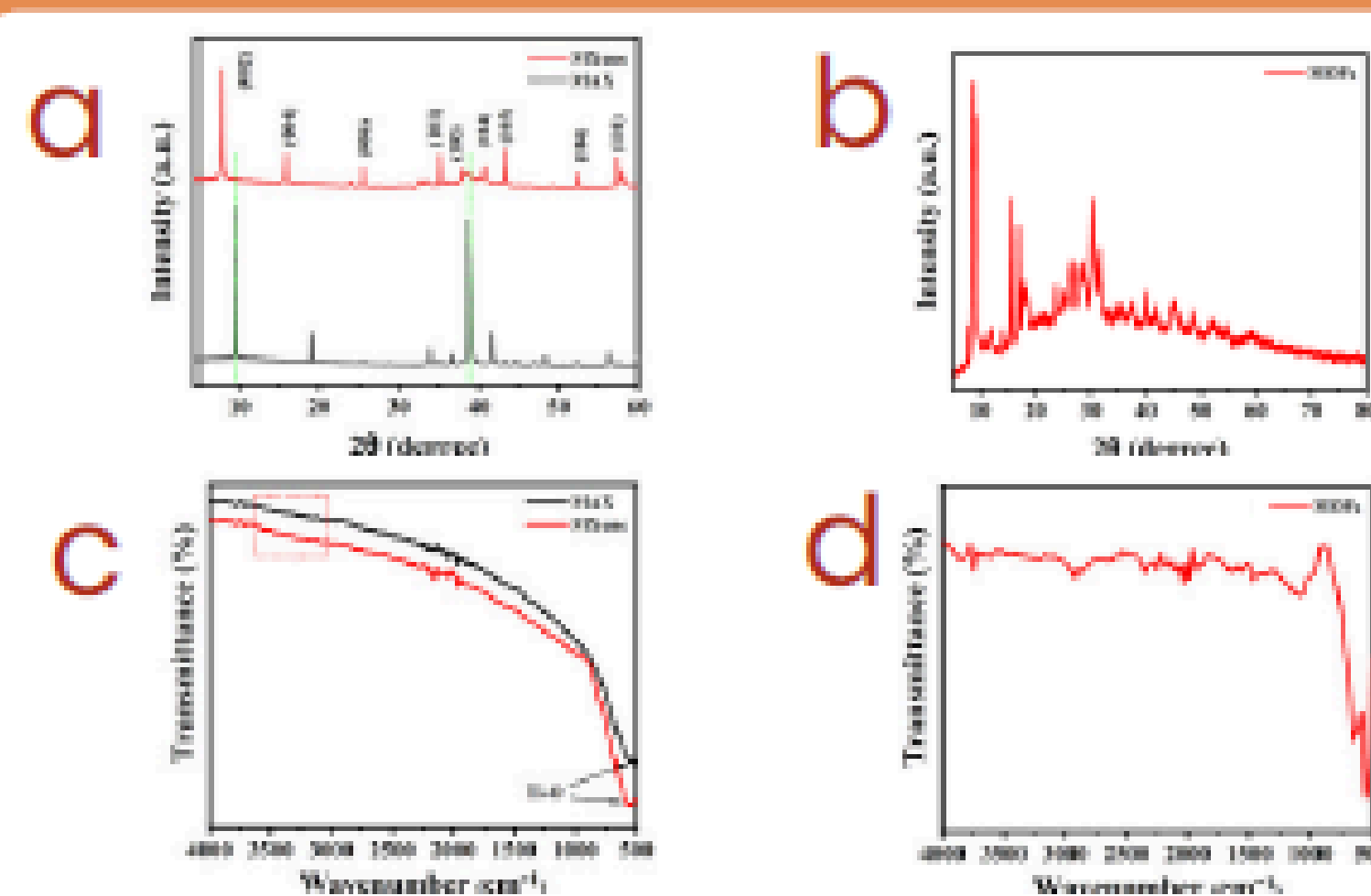


Fig. 2: Material Characterization (a) XRD confirms successful MAX-to-MXene conversion. (b) MOF shows crystalline structure. (c) FTIR verifies MXene formation via Ti-O bonds. (d) MOF surface groups support further hybridization.

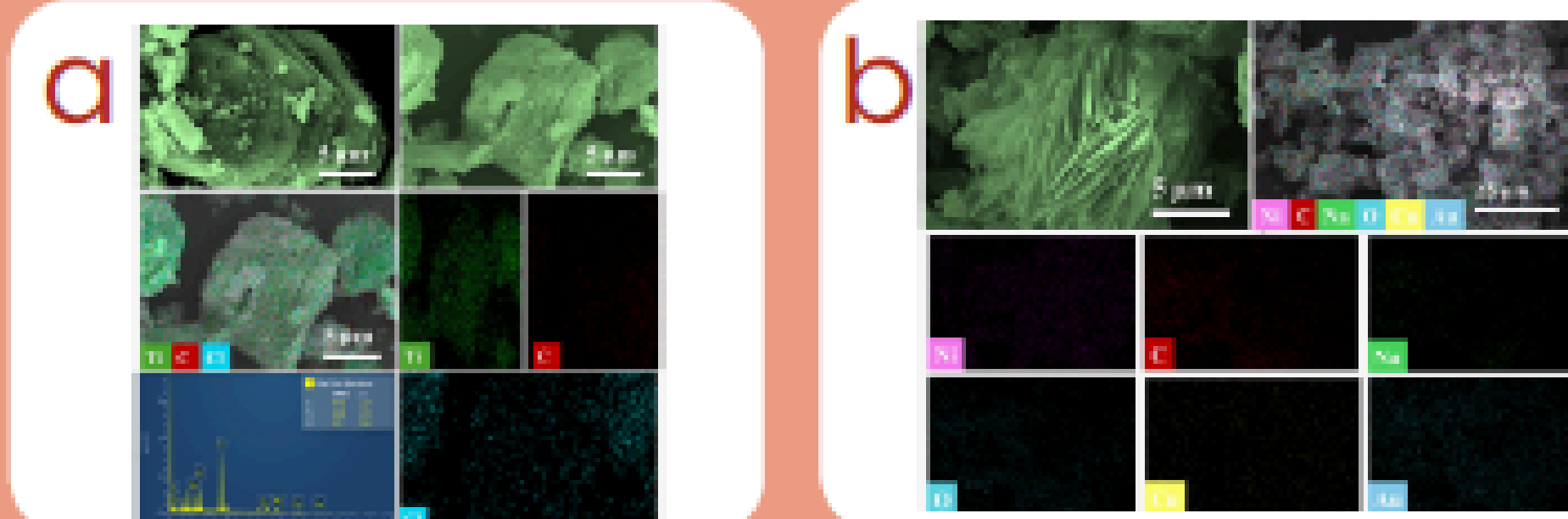


Fig. 3: SEM and Elemental Mapping (a) MXene exhibits a layered morphology with uniform distribution of Ti, C, and Cl, confirming successful etching and structural integrity. (b) MOF shows a flake-like structure with homogeneous dispersion of Ni, C, Na, O, Cu, and Au, indicating effective metal-ligand incorporation.

7 Results

Clean Etching

Active Surface

Uniform Structure

Characterization confirmed successful etching of MAX to MXene and MOF synthesis, with **active surface groups** and **uniform structure** verified by XRD, FTIR, SEM, and EDS analyses.

8 Conclusion

This work shows a **safer, fluoride-free** MXene-MOF synthesis, producing a **high-performance** material for flexible **sodium-metal batteries**. Its excellent structural and chemical properties highlight its potential for next-generation energy storage applications.

9 Application

Flexible/Wearable batteries

Sodium Metal Batteries

Electric Cars

Industrial Backup Power

Energy Storage

Portable Electronics

10 Future Work

Explore Alternative MAX Phases

Hybridize with Emerging MOFs

Introduce Dopants into the Polymer Electrolyte Matrix

11 References

